

List of Design Patterns

1. Creational Patterns
 - a. Factory Method
 - b. Abstract Factory
 - c. Builder Pattern
2. Structural Patterns
 - a. Adapter Pattern
 - b. Proxy Pattern
 - c. Flyweight Pattern
3. Behavioral Patterns
 - a. Strategy
 - b. Template
 - c. Observer
 - d. Mediator
 - e. Chain of Responsibility

Sample Questions

Q1: Design a Document Management System that can create different types of documents, such as PDF, Word, and Excel files.

Instructions:

- Create a common interface Document.
- Implement concrete document classes (PDFDocument, WordDocument, ExcelDocument).
- Create a factory method that generates the appropriate document type based on user selection.
- Demonstrate creating and opening different document types without directly instantiating their classes.

Q2: Design an Online Payment System that supports multiple payment methods such as Credit Card, PayPal, and Mobile Banking.

Instructions:

- Define a common payment strategy interface.
- Implement separate strategies for each payment method.
- The user should be able to select or change the payment method at runtime.
- Demonstrate processing payments using different strategies.